

Assessment of Alzheimer's Disease Progression Using Comorbidity-Aware Meta-Bayesian Framework with Resting-State fMRI

Second Review

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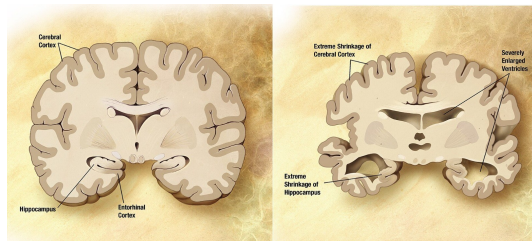
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Academic Year 2026–2027



Background and Motivation

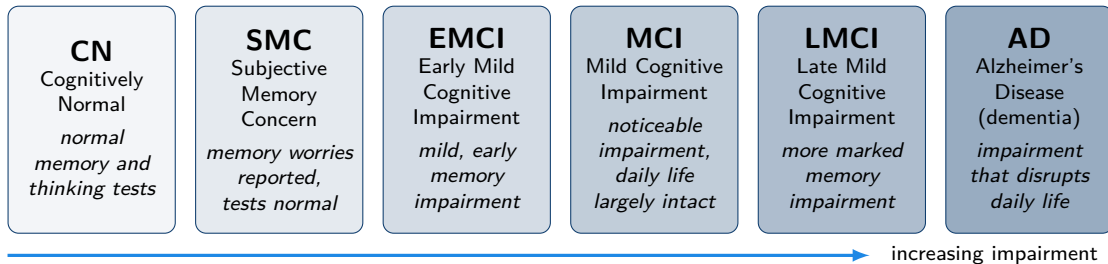
- ▶ **Alzheimer's disease (AD)** is the most common cause of dementia, about 60–70% of cases worldwide.
- ▶ It is **progressive**: memory, reasoning and daily function decline, mostly after age 65.
- ▶ India's ageing population is making the burden grow.
- ▶ Tissue loss (right) appears late. **Brain activity**, measured with functional MRI, can change earlier.



Healthy brain (left) and advanced AD (right): thinner cortex, shrunken hippocampus, enlarged ventricles.

Illustration reused from Review 1.

The Six Stages of Progression



- ▶ Labels are ADNI's diagnostic groups. Each stage has **about 30 scans**.
- ▶ The model treats the stages as an **ordered scale** (fixed at Review I): CN → SMC → EMCI → MCI → LMCI → AD.
- ▶ Neighbouring stages overlap, so every label should come with a **confidence**.

Problem Statement

Problem

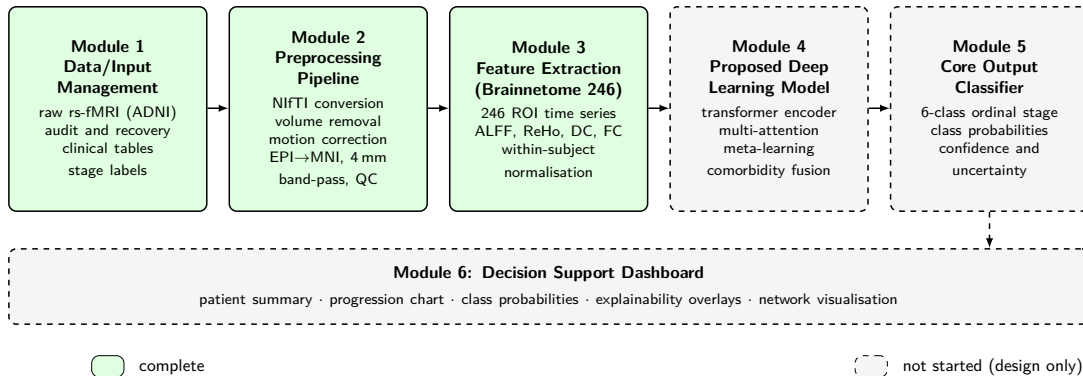
Existing systems analyse rs-fMRI biomarkers **independently** and omit clinical context, giving **overconfident, deterministic staging errors** on borderline subjects.

- ▶ **Context:** dementia progresses through six ordered stages.
- ▶ **Limitation:** pipelines isolate ALFF, ReHo, DC and FC.
- ▶ **Clinical gap:** models ignore how comorbidities modify neurodegeneration.

Research question

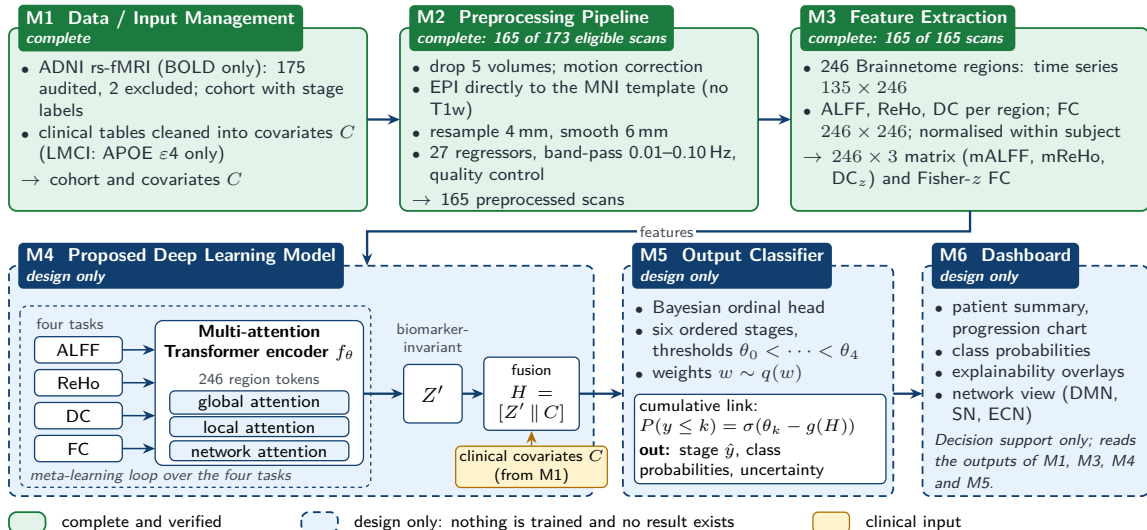
Can a common AD progression representation be found across several functional biomarkers, accounting for comorbidity and prediction uncertainty?

Proposed System Architecture



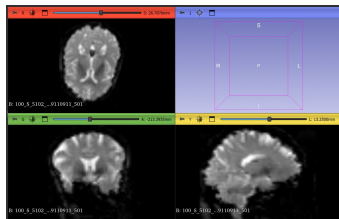
Goal: place each person on the six-stage scale from a resting-state scan and clinical context, and say how sure the model is.

Full System Architecture



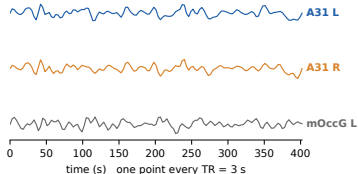
Resting-State fMRI in Plain Words

- ▶ The participant lies still in the scanner and does **no task**.
- ▶ Every 3 s the scanner takes a 3-D picture of the brain: 140 pictures, about 7 minutes.
- ▶ The signal is **BOLD**: local blood oxygenation, a proxy for neural activity.
- ▶ Slow fluctuations (0.01–0.10 Hz) in linked regions rise and fall together.



One converted scan in three planes (Review I).

left-right pair A31: $r = 0.90$ A31 L vs mOccG L: $r = -0.20$



Three regions from one scan: left and right A31 move together, an occipital region does not.

The Dataset: ADNI

- ▶ **ADNI** is a public, multi-site study that shares brain scans and clinical data [1, 2].
- ▶ We use its **resting-state fMRI only**: the download has no structural (T1) scans.
- ▶ Each scan: 140 volumes, $TR = 3$ s for most, voxels of about 3 mm.
- ▶ 175 scans audited, 2 excluded (1 duplicate, 1 truncated): 173 eligible, **165 processed**. Eight failed: their TR of 6.02 s is too slow for the 0.01–0.10 Hz band.
- ▶ **Target: 30 scans per stage, 180 in all.** The last column shows the processed scans still to add (SMC and MCI are already above 30).

Stage	Elig.	Proc.	To 30
CN	29	27	+3
SMC	32	31	0
EMCI	30	29	+1
MCI	35	32	0
LMCI	22	21	+9
AD	25	25	+5
Total	173	165	+18

Counts are per scan. Elig.: passed the audit; Proc.: preprocessed; To 30: processed scans still to add for 30 per stage.

Clinical Context: Covariates and Comorbidity

Covariate	What it is, and why it matters
Age	years at the scan; risk rises with age
APOE $\epsilon 4$	copies of the $\epsilon 4$ variant; strongest common genetic risk factor
MMSE	0–30 cognitive screening score; lower is worse
CDR-SB	0–18 clinician-rated severity; higher is worse
GDS	0–15 depression scale; often accompanies decline
Cardiovascular history	ADNI category MH4CARD; includes hypertension
Endocrine/metabolic history	ADNI category MH9ENDO; includes diabetes

What Is New in Our Approach

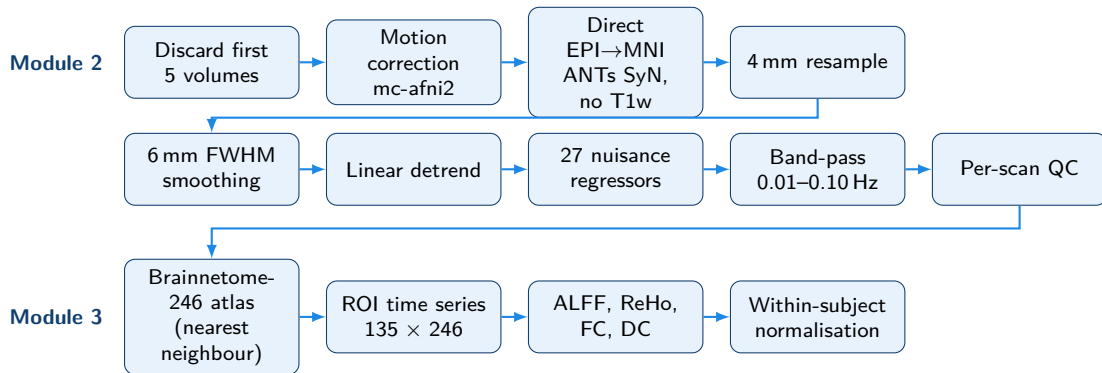
	Gap in existing work	What we propose (design)
G1	Each biomarker is analysed on its own	Meta-learning: each biomarker is a task; learn a shared, biomarker-invariant representation
G2	Clinical context is left out	Fusion: join covariates to the imaging representation, $H = [Z' \parallel C]$
G3	Progression treated as unordered classes	Ordinal model over six ordered stages
G4	Overconfident single labels	Bayesian output: probability per stage with uncertainty

Honest scope

All four are **design only**: nothing is trained. Of three 2026 papers checked, at least one partly addresses each of G2–G4 and none addresses G1 (report §1.3).

System Modules

Module (Review I names)	Status	At Review II
1 Data/Input Management	Complete	Imaging audited; clinical table cleaned (covariates for 107 of 127 subjects)
2 Preprocessing Pipeline	Complete	165 of 173 scans; 8 rejected (TR 6.02 s)
3 Feature Extraction (Brainnetome 246)	Complete	ALFF, ReHo, DC, FC for 165/165; normalised
4 Proposed Deep Learning Model	Not started	Transformer, meta-learning, fusion: design only
5 Core Output Classifier	Not started	Bayesian ordinal head: design only
6 Decision Support Dashboard	Not started	No interface built



Every step is checked per acquisition; a scan that fails a check is logged, never silently altered.

Preprocessing: What Each Step Does and Why

Step	What it does	Why it is needed
Discard first 5 volumes	drops the first 15 s	the signal is still settling at the start
Motion correction	aligns every volume to the first	head movement would otherwise look like brain activity [8, 9]
Warp to MNI template	maps each brain onto a standard brain [10]	so region k is the same place in every person
Resample to 4 mm, smooth 6 mm	coarser grid and slight blur	less noise and fewer voxels
Detrend and 27 regressors	removes slow drift and motion, white-matter and whole-brain signal	non-neural influences hide the neural signal
Band-pass 0.01–0.10 Hz	keeps only slow fluctuations	the range where resting-state activity lives
Quality control	motion, signal-to-noise and file checks per scan	failures are logged, never silently altered

The Four Feature Biomarkers

ALFF

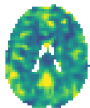
how active



How big the ups and downs are in one region.
Bigger: more active.

ReHo

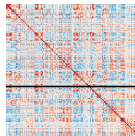
teamwork inside



Do the tiny spots inside one region go up and down together?
Yes: it works as a team.

FC

who goes with whom



Do two regions go up and down at the same time?
Yes: they are connected.

DC

how well linked



Add up a region's links to all the other regions.
A big total: well linked.

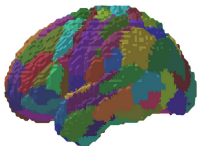
Think of a brain region as a group of people clapping

ALFF: how loud they clap. **ReHo:** do they clap together?

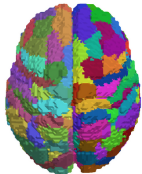
FC: does another group clap at the same moments? **DC:** how many groups clap along with it?

The Brain Atlas: 246 Regions

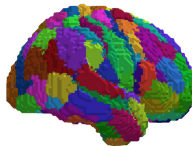
Left hemisphere



Top view



Right hemisphere



Sagittal slice



Coronal slice



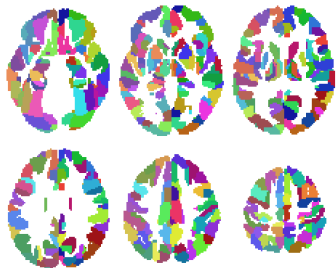
Axial slice



- ▶ An **atlas** divides the brain into named regions so scans can be compared region by region.
- ▶ **Brainnetome atlas**: 246 regions, 123 per hemisphere [3]. Each colour is one region.
- ▶ Every scan is warped to a common template (MNI space), so region k is the same place in every person.
- ▶ A region's signal is the average of its voxels: **246 time series per scan**.

Brainnetome 246-ROI Parcellation for Feature Extraction

Fine-grained, whole-brain representation for systematic regional and inter-regional feature extraction



Axial slices, $z = -16$ to $+56$ mm.

Each colour is one ROI.

- ▶ **Fine-grained whole-brain coverage.** 246 anatomically defined ROIs covering cortical and subcortical regions.
- ▶ **Complete extraction space.** 246/246 ROIs present across all 165 acquisitions; no empty or zero-variance ROIs.
- ▶ **Systematic feature extraction.** Each ROI provides regional biomarkers (ALFF, ReHo, DC) and participates in the 246×246 FC matrix.
- ▶ **No premature feature selection.** All ROIs are retained to preserve spatial information; predictive importance will be learned by the downstream model.

246

ROIs

165

acquisitions

135×246

ROI time series per scan

246×246

FC matrix per scan

The 246 ROIs are not assumed to be equally disease-relevant; they define a complete, reproducible feature-extraction space from which the model will learn predictive regional and connectivity patterns.

Core Algorithm A: Implemented Biomarkers and Normalisation

Biomarkers (per Brainnetome region)

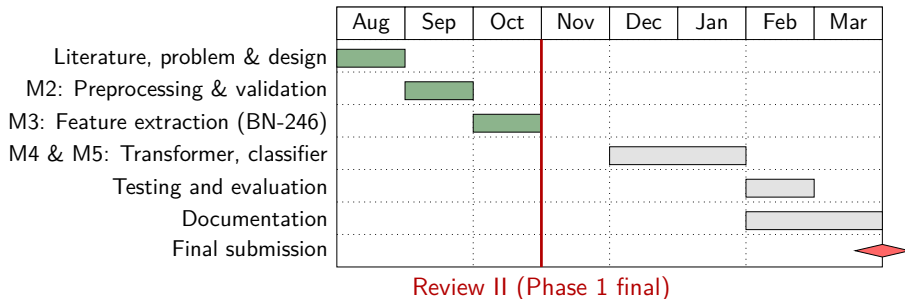
- ▶ **ALFF**: mean FFT amplitude, 0.01–0.10 Hz
- ▶ **ReHo**: Kendall's W , 27-voxel neighbourhood
- ▶ **FC**: 246×246 Pearson, signed, unthresholded
- ▶ **DC**: $DC_i = \sum_{j \neq i} FC_{ij}$

Within-subject normalisation

- ▶ $mALFF = ALFF_i / \overline{ALFF}$
- ▶ $mReHo = ReHo_i / \overline{ReHo}$
- ▶ $DC_z = (DC_i - \overline{DC}) / \sigma_{DC}$
- ▶ $FC \rightarrow \text{artanh}(\text{clip}(r, \pm 0.999999))$, diagonal = 0

- ▶ Every statistic comes from **one acquisition's own 246 values**.
- ▶ No cross-subject statistic and no label: leakage-free by construction.
- ▶ Verified: $\text{mean}(mALFF) = \text{mean}(mReHo) = 1.000000$; $\text{std}(DC_z) = 1.000000$ (165/165).
- ▶ 0 off-diagonal entries needed clipping; Fisher-z diagonal exactly 0.

Implementation Status



One milestone ahead of the Review I plan

Module 2 was planned for Review II and Module 3 for Review III. **Both are complete at Review II**, the final review of Phase 1.

Applies to the **data foundation only** (Modules 1–3); Modules 4–6 are not started. Module 2 completed in September; Module 3 in October (normalised biomarkers 5 Oct).

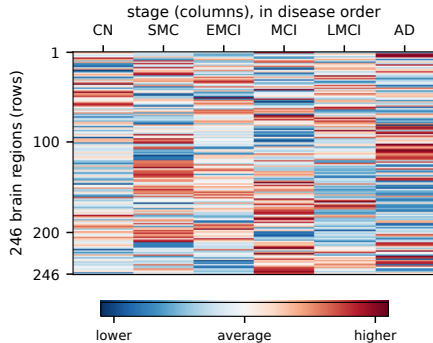
Results: Data Foundation Verification

These are **data-stage** results. **No model performance exists.**

Check (165 acquisitions)	Result
246/246 regions covered	165/165
ROI series (135, 246)	165/165
NaN / Inf / zero-variance	0 / 0 / 0
Pilot reproduction	bit-identical
DC recomputed from FC, max diff.	0.000e+00
Consistency checks passed	165/165

- ▶ 127 subjects, not 165 scans: 96 have one scan, 25 two, 5 three, 1 four.
- ▶ Smallest class: MCI, 32 scans but only **14 subjects**.
- ▶ QC medians ($n = 164$, one artefact scan excluded): tSNR 127.95, SNR 13.12, FD 0.317 mm.

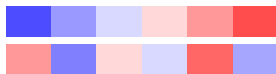
How to Read the Stage Heatmaps



mALFF, 127 subjects (real data).

- ▶ Each **row** is a brain region (246), each **column** a stage. A cell is the group average for that stage.
- ▶ **Colour** is relative to the region's own average over the six stages (row-wise z-score): absolute levels are removed on purpose.
- ▶ Disease-driven change would appear as a row shifting **steadily** from one colour to the other, left to right.
- ▶ **By chance**, an unrelated region is steadily ordered with probability $2/6! \approx 0.28\%$: about **0.7 of 246 regions**. We found 1, 0 and 1.

CN SMC EMCI MCI LMCI AD



the pattern we looked for

a typical region

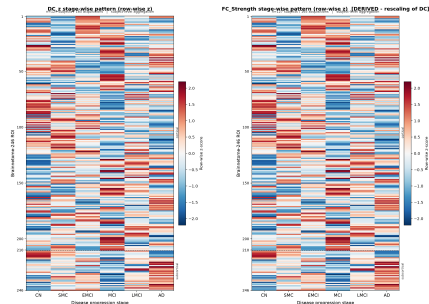
schematic, not data

Finding 1: FC Strength Equals DC/245

Review I fourth column was redundant

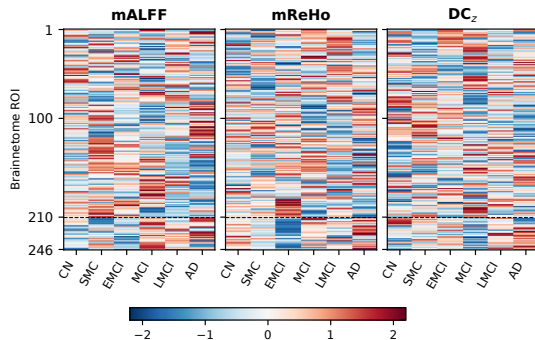
$$FCS_i = \frac{1}{245} \sum_{j \neq i} FC_{ij} = \frac{DC_i}{245}$$

- ▶ Max difference **6.939e-18**; Pearson $r = 1.000000000000$.
- ▶ **All four biomarkers are kept.**
- ▶ ALFF, ReHo, DC: 246×3 regional matrix.
- ▶ FC: full 246×246 Fisher-z matrix.
- ▶ A design correction found by auditing our own features.



DC_z (left) and FC_Strength (right): identical panels, as expected. Descriptive and exploratory only.

Finding 2: No Univariate Progression Signal



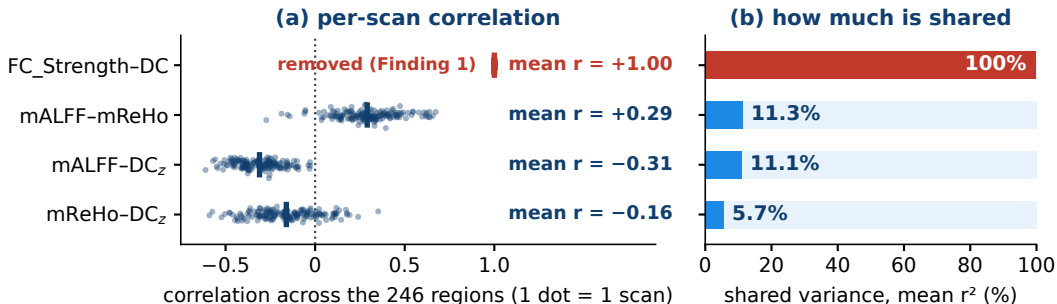
Stage-wise row-z patterns, 127 subjects (CN→AD).

Descriptive and exploratory; no statistical test.

	Monotonic CN→AD	By chance
mALFF	1 / 246	≈ 0.7
mReHo	0 / 246	≈ 0.7
DC _z	1 / 246	≈ 0.7

- ▶ No more than chance; stage variation is 4–5× below anatomy.
- ▶ Informative, not a failure.
- ▶ Group means discard the whole connectivity structure.
- ▶ **Motivates the multivariate model.**

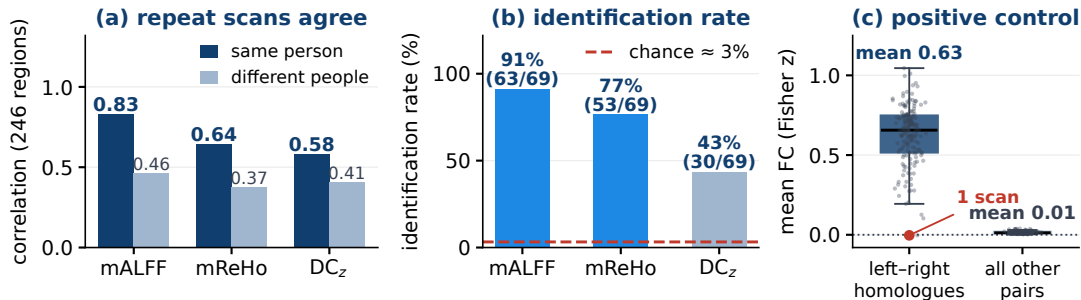
Finding 3: The Biomarkers Are Not Redundant



FC strength equals DC/245 exactly (Finding 1) and was removed. The three kept biomarkers share only **6–11 %** of their variance across the 246 regions (**89–94 % non-redundant**, all 165 scans), and mALFF and DC_z are negatively related in every scan.

Gap G1 uses each biomarker as a separate task. That only makes sense if they carry different information, and they overlap little. Whether it improves staging is untested (Phase 2).

Finding 4: The Features Are Reproducible and Plausible



31 people have repeat scans (69 scans; 45 of 46 pairs are from different sessions). For mALFF, a scan's closest match is the same person's other scan in **63 of 69** cases (chance $\approx 3\%$), and left-right homologous regions are strongly connected in **164 of 165** scans (one EMCI scan lacks this and needs checking).

Data checks, not a disease classifier: no stage label is used. A person's repeat scans usually share a site and scanner, so part of the reproducibility may be scanner-related.

Team Responsibilities

Team member	Responsibility	Status at Review II
Harishkanna R	Preprocessing (Module 2); transformer model (Module 4)	Preprocessing: Complete Transformer: Not started
Dhananchezhiyan M	Meta-Bayesian module (meta-learning, Bayesian classifier) and final module (dashboard)	Not started
Nagindhar Sachein R L	Dataset cleaning (clinical and comorbidity tables, dataset expansion)	Clinical cleaning done and audited; dataset expansion next

Feature extraction (Module 3) was completed by the team and is verified.

References

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- [18] B.-S. Rauchmann *et al.*, "Multimodal ordinal modeling of AD severity (MRI and clinical)," arXiv:2606.11794, 2026.
- [19] K. Alsalem *et al.*, "Progression-aware CNN-Transformer for multiclass AD staging," *Diagnostics*, 16(4):593, 2026.

Thank You

Questions and Discussion

Backup: Glossary

Term	Meaning	Term	Meaning
rs-fMRI	functional MRI recorded while the person rests	ALFF	amplitude of low-frequency fluctuations: size of the slow signal swings
BOLD	blood-oxygen-level-dependent signal measured by fMRI	ReHo	regional homogeneity: how in step a region's voxels are
TR	time between two brain pictures (3 s here)	FC	functional connectivity: correlation of two regions' signals
Voxel	one 3-D pixel of a scan (a 4 mm cube after resampling)	DC	degree centrality: a region's total connectivity (sum of its FC row)
ROI, atlas	region of interest; a map that divides the brain into regions	ADNI	Alzheimer's Disease Neuroimaging Initiative
MNI space	standard template brain that all scans are aligned to	Ordinal	classes with a natural order (CN < ... < AD)
Band-pass	keep only 0.01–0.10 Hz	Meta-learning	learning, across several tasks, what they have in common
Nuisance regression	removing motion and other non-neural signals	Bayesian model	outputs a distribution over predictions, giving uncertainty
Row-wise z	a row rescaled by its own mean and spread	Grouped CV	validation that keeps all scans of one person in the same fold

Backup: Data Limitations

- ▶ **No structural (T1) scans:** each scan was registered directly to the template, with no subject-specific tissue masks, so no CSF regressor: 27 regressors (24 motion terms, white matter, global signal).
- ▶ **Slice-timing correction not applied:** the metadata needed to recover the slice order are not available for this dataset, and nothing was inferred.
- ▶ **Eight scans rejected:** $TR \approx 6.02\text{ s}$ makes the 0.10 Hz band-pass impossible (Nyquist).
- ▶ **No cross-site harmonisation:** ADNI is multi-site, so scanner effects may remain.
- ▶ **127 subjects, 14 in the smallest class (MCI):** model capacity is limited by subjects, not scans; validation must keep each person's scans together.
- ▶ **LMCI covariates missing:** APOE $\epsilon 4$ only for the 20 LMCI subjects; 116 duplicate conflicts in the clinical sources await review.
- ▶ **No univariate stage signal** in the group means (Finding 2): the case for a multivariate model.

Backup: Residual Motion and Site Effects

Motion survives the 27 regressors

Feature	Regions linked to FD
mALFF	129 / 246
mReHo	131 / 246
DC _z	39 / 246
expected by chance	≈ 12

Pearson r between a region's value and each scan's mean FD, $p < 0.05$ uncorrected, 164 scans (artefact scan excluded). Repeat scans of one person are not independent.

- ▶ Mean FD will enter the model as a covariate. DC_z is less affected, but still above chance.

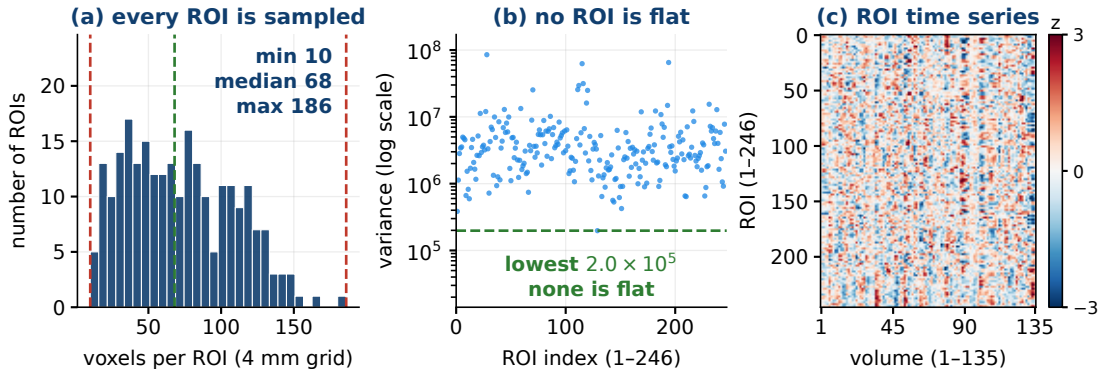
Site is concentrated where it hurts

	Scans	Sites	Top site	Top two
CN	27	11	19 %	37 %
SMC	31	10	19 %	32 %
EMCI	29	7	21 %	41 %
MCI	32	8	31 %	47 %
LMCI	21	7	29 %	48 %
AD	25	7	40 %	68 %

Share of a stage's scans from its largest and two largest sites (site = prefix of the ADNI subject ID); 14 sites in all.

- ▶ Any CN–AD contrast is partly a site contrast: harmonisation or a site covariate is needed before Module 4.

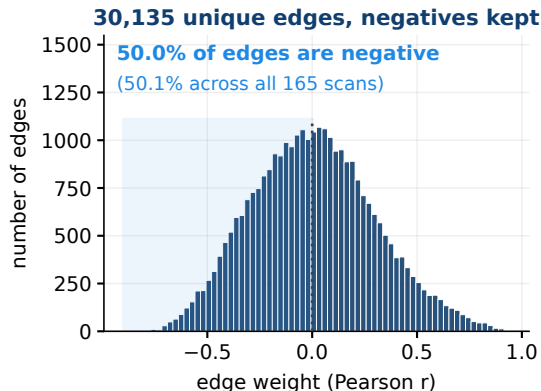
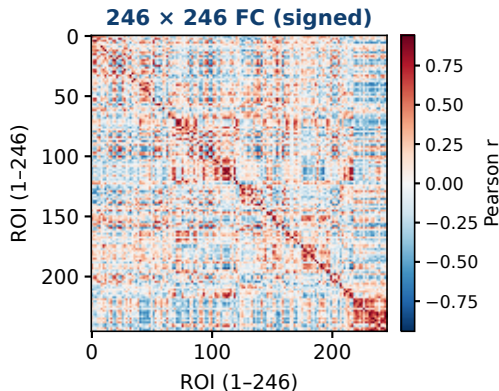
Backup: The 246-ROI Extraction Space Is Verified



246/246 ROIs are present in all 165 acquisitions: no empty ROI (the smallest has 10 voxels), no zero-variance time series, no NaN or Inf. Every scan gives the same 135×246 matrix.

(a): the atlas on the 4 mm grid. (b), (c): one representative CN scan; (c) is z-scored per ROI for display.

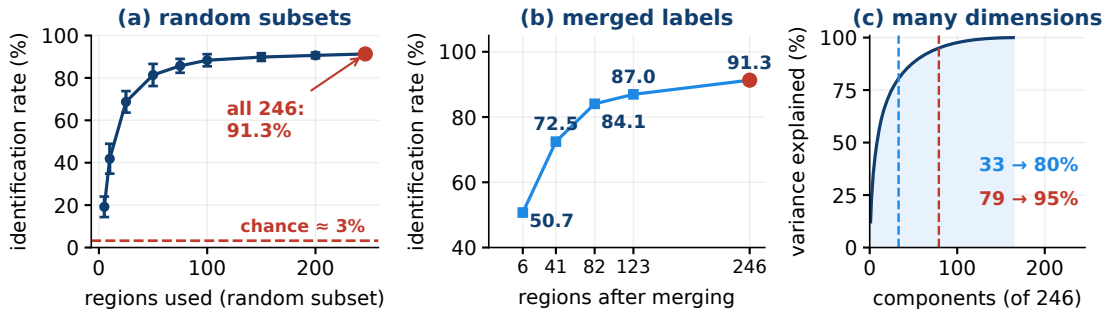
Backup: 246 Regions as Connectivity Nodes



Signed and unthresholded: all 30,135 unique edges are kept for each of the 165 acquisitions. About half are negative (50.1 % over the 165 scans; per-scan range 45.6–53.1 %), as expected after global signal regression, which is part of the 27-regressor design and centres the values near zero.

Matrix and histogram: one representative CN scan.

Backup: Why 246 Regions?



Closest-match identification (mALFF; 69 scans, 31 people; no stage label): merging consecutive atlas labels loses information at every step, but returns flatten (a random 100 regions: 88.3%; all 246: **91.3 %**). Two regions share only about 4% of their variance (165 scans).

Finer regions keep information that coarser versions of the same data lose. It does not show that 246 is optimal for staging disease: only the trained model can test that.

Backup: Technology Stack

Used so far (data foundation)

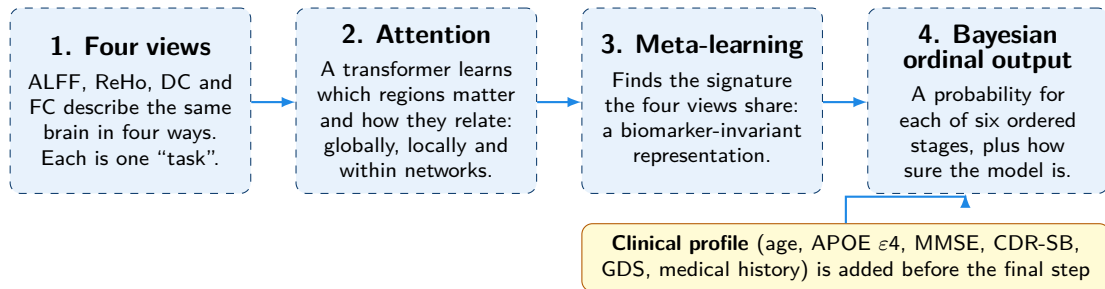
- ▶ Python 3.12.3, NumPy, SciPy, pandas
- ▶ NiBabel 5.4.2, Nilearn 0.14.1
- ▶ ANTsPy 0.6.3 (registration), TemplateFlow 25.1.2
- ▶ FreeSurfer 7.4.1 (FS-FAST motion correction)
- ▶ WSL2 (Ubuntu) on Windows 11; Git

Planned (not yet used)

- ▶ PyTorch, GPyTorch (model, Bayesian layers)
- ▶ Streamlit (Module 6 dashboard)
- ▶ Hardware per Review I: Intel Core i7, 32 GB RAM, RTX 4060 (8 GB)

Dataset: ADNI via LONI. Versions are those recorded in each acquisition's provenance file.

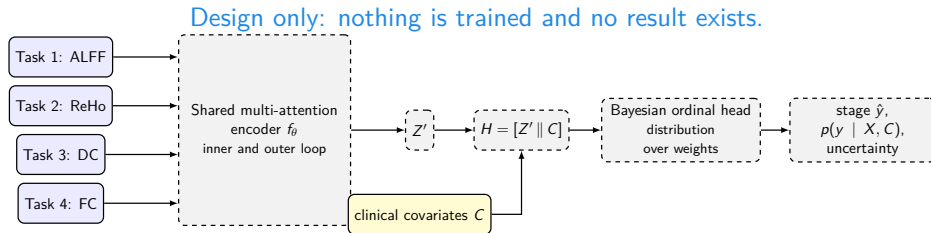
The Meta-Bayesian Model in Plain Words (Design)



- ▶ **Ordinal:** CN→AD is a progression, so confusing neighbouring stages is a smaller error than confusing distant ones.
- ▶ **Bayesian:** borderline subjects get a wide uncertainty instead of a falsely confident label.

Design only. Nothing is trained; no accuracy, calibration or attention result exists.

Core Algorithm B: Meta-Bayesian Pipeline (Design, Not Implemented)



MAML update over biomarkers

$$\theta'_b = \theta - \alpha \nabla_{\theta} \mathcal{L}_b^{\text{sup}}(\theta)$$
$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_b \mathcal{L}_b^{\text{qry}}(\theta'_b)$$

Ordinal head (cumulative link)

$$P(y \leq k | H, w) = \sigma(\tau_k - g_w(H)), \quad \tau_0 < \dots < \tau_4$$

Support and query sets are subject-disjoint, within training folds. Uncertainty = predictive entropy over S weight samples.

First Review - Queries and Answers

Panel comment	Response	Detail in
Need more clarity on the dataset (biomarkers)	175 audited, 165 preprocessed, 127 subjects. Four biomarkers on 246 regions.	Dataset and biomarker slides; report §6.1
Need more clarity on the Meta-Bayesian framework	“Meta”: biomarkers as tasks. “Bayesian”: ordinal head with uncertainty. Design only.	Phase 2 slides (at the end); report §4.5
Need more clarity on the comorbidity data	Seven covariates, listed on the covariates slide. Cleaned table for 127 subjects; the LMCI group has APOE only.	Covariates slide; report §4.5